

■ 3.5 Calculus

■ 3.5.1 Differentiation

$D[f, x]$	partial derivative $\frac{\partial}{\partial x} f$
$D[f, x_1, x_2, \dots]$	multiple derivative $\frac{\partial}{\partial x_1} \frac{\partial}{\partial x_2} \dots f$
$D[f, \{x, n\}]$	n^{th} derivative $\frac{\partial^n}{\partial x^n} f$
$D[f, x, \text{NonConstants} \rightarrow \{v_1, v_2, \dots\}]$	$\frac{\partial}{\partial x} f$ with the v_i taken to depend on x

Partial differentiation operations.

This gives $\frac{\partial}{\partial x} x^n$.

```
In[1]:= D[x^n, x]
Out[1]= n x-1 + n
```

This gives the third derivative.

```
In[2]:= D[x^n, {x, 3}]
Out[2]= n x-3 + n (-2 + n) (-1 + n)
```

You can differentiate with respect to any expression.

```
In[3]:= D[ x[1]^2 + x[2]^2, x[1] ]
Out[3]= 2 x[1]
```

D does *partial differentiation*. It assumes here that y is independent of x .

```
In[4]:= D[x^2 + y^2, x]
Out[4]= 2 x
```

If y does in fact depend on x , you can use the explicit functional form $y[x]$. Section 3.5.4 describes how objects like $y'[x]$ work.

```
In[5]:= D[x^2 + y[x]^2, x]
Out[5]= 2 x + 2 y[x] y'[x]
```

Instead of giving an explicit function $y[x]$, you can tell D that y *implicitly* depends on x . $D[y, x, \text{NonConstants} \rightarrow \{y\}]$ then represents $\frac{\partial y}{\partial x}$, with y implicitly depending on x .

```
In[6]:= D[x^2 + y^2, x, NonConstants -> {y}]
Out[6]= 2 x + 2 y D[y, x, NonConstants -> {y}]
```